

The 7 Steps of Data Analysis

A clear, practical guide to turning raw data into smart business decisions

STEP 01 Define Question	STEP 02 Collect Data	STEP 03 Inspect Data	STEP 04 Clean Data	STEP 05 Explore (EDA)	STEP 06 Find Answers	STEP 07 Share & Act
Start with the real business problem you need to solve. - Ask why it matters - Set success metrics	Gather relevant files, numbers, and survey records. - Locate data sources - Save safe raw copy	Look over your data to understand rows and columns. - Check empty blanks - Catch obvious errors	Fix typos, fill empty cells, and remove duplicates. - Remove duplicates - Standardize formats	Make simple charts to discover patterns and trends. - Draw bar & line charts - Spot surprising trends	Compare groups and find what truly caused the results. - Compare segments - Test real drivers	Tell the story clearly to help leaders make smart decisions. - Simple visual story - Clear next steps
DELIVERABLE Problem Statement	DELIVERABLE Raw Data Backup	DELIVERABLE Data Checklist	DELIVERABLE Clean Dataset	DELIVERABLE Visual Charts	DELIVERABLE Key Findings	DELIVERABLE Action Plan

KEY PRINCIPLE

Data Analysis is an Iterative Loop: If your charts look unexpected, you can always go back and inspect your data.

↶ Clean → Inspect

↶ Charts → Question

Step 1: Define the Problem

Start with a clear business question before touching any spreadsheet

What to Do in Step 1

Step 01 // Foundation

1

Ask a Specific Business Question
State your problem in plain language (e.g., "Why did sales drop last month?" or "Which customers leave first?"). Avoid vague questions like "Tell me about our data."

2

Know Who Will Read the Answer
Identify your audience: Is it your manager, marketing team, or company executive? Knowing who will read the answer shapes what you focus on.

3

Pick 1 or 2 Clear Success Metrics
Choose measurable target numbers upfront (e.g., total sales revenue, customer return rate, or average delivery time) so you know when you succeed.

★ PRIMARY DELIVERABLE

One-Page Problem Statement: A simple document stating the exact question, why it matters, who needs the answer, and what metric defines success.

💡 BEGINNER GOLDEN RULE

Never touch data without a question: If you start digging into spreadsheets before defining your goal, you will get lost in random numbers and waste hours.

EVERYDAY EXAMPLES

Problem Statement

KPI Checklist

Target Audience

RULE OF THUMB

Start with the Question: A clear question answered with simple math is worth more than complex algorithms answering the wrong question.

Step 2: Collect the Data

Gathering the right numbers, tables, and records to answer your question

What to Do in Step 2

Step 02 // Gathering

1

Find Where Your Data Lives
Check your company databases, Excel files, Google Forms, survey results, or public websites. Ask team members where the official records are kept.

2

Collect Only What is Relevant
Don't try to download everything. Gather only the specific columns, dates, and tables needed to answer your Step 1 question.

3

Save an Untouched Original Copy
Immediately save an untouched, read-only backup file before making any changes. Always do your working analysis on a separate copy.

★ PRIMARY DELIVERABLE

Untouched Raw Data Backup: An archived original data file with the date, source location, and download notes safely recorded.

💡 BEGINNER GOLDEN RULE

Always preserve the original: If you accidentally delete a column or overwrite values during cleaning, your raw backup is your only safety net.

COMMON DATA FORMATS

CSV Files

Excel (.xlsx)

Google Sheets

SQL Tables

DATA SAFETY

Rule of Duplication: Never edit your only raw data file. Always duplicate first, then analyze safely.

Step 3: Inspect the Data

Looking closely at your dataset before running any calculations or charts

What to Do in Step 3

Step 03 // First Look

1

Check Table Size & Column Names
Look at how many rows and columns you have. Read each header name to make sure you understand what each column represents.

2

Look for Missing Empty Cells
Find which columns have missing blanks. Are customers leaving phone numbers empty? Are product prices missing?

3

Spot Obvious Nonsense Numbers
Scan for obvious red flags: negative prices, an age of 200, future dates on past orders, or test accounts like "test@test.com".

★ PRIMARY DELIVERABLE

Data Health Checklist: A quick summary listing total row counts, column types, missing value counts, and suspicious values spotted.

💡 BEGINNER GOLDEN RULE

Inspect before calculating: Spending 10 minutes looking over the raw rows prevents hours of recalculating flawed results later.

QUICK INSPECTION CHECKS

Row / Column Count

Min & Max Values

Blank Cell Count

EYE TEST

The 10–Minute Sanity Check: If numbers look suspicious at first glance, stop immediately and investigate before calculating.

Step 4: Clean the Data

Fixing errors, blanks, and typos so your numbers tell the truth

What to Do in Step 4

Step 04 // Cleanup

1

Remove Duplicate Rows
Find and delete accidental duplicate rows so single transactions or customer signups are never counted multiple times.

2

Handle Empty Cells Wisely
Decide whether to fill empty cells with logical defaults (like zero or the column average) or remove rows that lack critical information.

3

Standardize Formats & Fix Typos
Convert inconsistent words like "USA", "U.S.", and "United States" into a single name. Make sure all dates use the same format (YYYY-MM-DD).

★ PRIMARY DELIVERABLE

Clean, Model-Ready Dataset: A standardized, duplicate-free spreadsheet where every column has valid formats and clear values.

💡 BEGINNER GOLDEN RULE

Garbage in, garbage out: 80% of data work is cleaning. If messy data enters your formulas, misleading conclusions come out.

CORE CLEANING TASKS

Remove Duplicates

Trim Extra Spaces

Standardize Dates

RE-CHECK LOOP

Clean → Inspect Again: After cleaning, quickly re-check your row counts to ensure no good records were erased.

Step 5: Explore the Data (EDA)

Using simple visuals to spot trends, patterns, and unexpected surprises

What to Do in Step 5

Step 05 // Discovery

1

Calculate Simple Summary Numbers
Find the average, middle value (median), minimum, and maximum for your key numbers to see what a "typical" record looks like.

2

Draw Simple, Intuitive Visuals
Use bar charts to compare categories, line charts to track trends over time, and scatter plots to see if two variables move together.

3

Look for Surprises & Trends
Notice unexpected patterns: Did orders spike on Friday nights? Do mobile shoppers spend more than desktop shoppers?

★ PRIMARY DELIVERABLE

Key Charts & Trend Summary: A curated set of 3 to 5 clear visual charts highlighting the most interesting patterns in your data.

💡 BEGINNER GOLDEN RULE

Let the charts tell the story: Don't try to prove your personal hunch. Look at what the visual charts actually show with an open mind.

THE 3 ESSENTIAL CHARTS

Bar (Compare groups)

Line (Over time)

Scatter (Relationships)

EXPLORATION

Keep Visuals Simple: If a chart requires a 5-minute explanation, simplify it into a clear bar or line chart.

Step 6: Analyze & Find Answers

Testing what really caused the outcome and summarizing your key findings

What to Do in Step 6

Step 06 // Analysis

1

Compare Groups to Test Ideas
Divide your data into groups (e.g., customers who received discounts vs. those who did not) to see if their spending was truly different.

2

Separate True Causes from Coincidence
Check if a relationship is real or just lucky timing. Remember: just because two trends rise together does not mean one caused the other.

3

Answer Your Step 1 Question
State 2 or 3 direct conclusions with concrete numbers (e.g., "Discount coupons increased total sales by 18%, but reduced profit margins by 4%").

★ PRIMARY DELIVERABLE

Core Findings & Evidence Summary: A concise summary of the key findings that directly answer your original business question.

💡 BEGINNER GOLDEN RULE

Correlation is not causation: Ice cream sales and swimming pool accidents both rise in summer, but ice cream does not cause accidents!

CORE ANALYSIS METHODS

A/B Comparisons

Percentage Growth

Key Driver Analysis

REAL ANSWERS

Numbers With Context: Never report raw metrics alone. Always explain what the numbers mean for the team.

Step 7: Share & Take Action

Turning data insights into a clear story that drives real business decisions

What to Do in Step 7

Step 07 // Value

1

Tell a Clear, Simple Story
Present your findings in straightforward language. Skip technical details and explain what the results mean for the team.

2

Recommend Concrete Next Steps
Suggest 1 or 2 specific actions based on your findings (e.g., "Target email discounts only to users inactive for 30+ days").

3

Track What Happens After
Monitor key metrics after your recommendations are launched to see if the changes actually produced the expected improvements.

★ PRIMARY DELIVERABLE

Action Plan & 1-Page Summary Deck: An executive presentation slide outlining the key finding, recommended action, and expected result.

💡 BEGINNER GOLDEN RULE

Action is the goal: The purpose of data analysis is not to show how hard you worked, but to help people make the right decision.

FINAL DELIVERABLE FORMATS

Executive Slide

Action Checklist

Live Dashboard

THE FINAL GOAL

Close the Loop: Great data analysis turns insights into actions, and actions into measurable results.